Security-Focused Fault Catalogue for a Cloud-Native 5G Core Dataset

Draft fault list for extending the BSc cross-layer fault atlas into a multimodal dataset (AnoMod/LO2 style). Sources: **Thesis** = Bonev BSc final paper; **Aiello** = Aiello et al., 5G Control/User Plane security survey (JCP 2026); **CVE/Open5GS** = public advisories & issue tracker; **Tool** = injection tooling. Compiled 23 Sep 2026.

Level A — Control-plane signaling attacks

ID	Fault	Target	How to simulate	Source / reference
A1	**Registration storm**	AMF, AUSF, UDM	Spawn 100s–1000s of UERANSIM UEs (loop over IMSIs) sending Registration Requests in a tight window.	**Aiello**: registration storms 0.80 (Critical). **Thesis**: fixes the admitted light control-plane workload. Method mirrors Springer 2026 'AMF/UPF saturation via virtual SIMs'.
A2	**NGAP signaling flood**	AMF (N2)	Replay/inject NGAP over SCTP port 38412 with 5GReplay (increase nb-copies), or scripted gNB repeating NGAP.	**Aiello**: NGAP floods 0.82 (Critical). **Tool**: 5GReplay (Montimage, ARES 2021) — hit OGS_CLUSTER error in Open5GS at ~1780 copies.
A3	**NAS malformed / auth-reject flood**	AMF, AUSF	UERANSIM UEs with bad keys/SUCI → repeated auth failures; or 5GReplay NAS mutation.	**Aiello**: 5G-AKA weaknesses 0.62 (High). **Thesis**: finally exercises amf_auth_fail / amf_auth_reject counters that never fired.
A4	**Paging / N2-setup abuse (optional)**	AMF	Repeated NG-Setup / paging-triggering signaling via crafted gNB.	**Aiello**: paging storm attack (Sec. 4.3). Lower priority; breadth only.

Level B — PFCP / user-plane protocol attacks

ID	Fault	Target	How to simulate	Source / reference
B1	**PFCP association flood**	SMF/UPF (N4)	pfcpsim: 'pfcpctl session create --count ...' + high-rate association setup against UPF.	**Aiello**: PFCP assoc. flooding 0.68 (High). **Tool**: pfcpsim (omec-project/ONF), containerised. Extends **Thesis** PFCP class from Chaos-Mesh proxy to real protocol.
B2	**Crafted / malformed PFCP flood**	UPF, SMF	pfcpsim fuzzing mode, or public PoC scripts; pin Open5GS version and report which still reproduce.	**CVE**: CVE-2024-51179, CVE-2025-29646, CVE-2026-92416; Open5GS issues #3497/#3498. A patched (non-firing) case is a valid negative result.
B3	**PFCP session deletion / hijack**	UPF	Crafted PFCP Delete Session Request with a sniffed SEID; GTP-U spoof + PDR redirect.	**Open5GS**: issues #3693, discussion #3919 (session hijack / DoS). **Aiello**: PFCP misuse listed among top risks.
B4	**GTP-U amplification / echo abuse**	UPF (N3/N9)	Spoofed GTP-U echo requests (scapy or 5GReplay on the data plane).	**Aiello**: GTP-U amplification 0.72 (High); notes user plane is under-studied.
B5	**Network-degradation baseline (keep 1)**	UPF	Chaos Mesh NetworkChaos loss/corrupt/duplicate, as in original thesis PFCP variants.	**Thesis**: original PFCP-attack class. Keep ONE to contrast 'protocol attack vs. equivalent packet loss'.

Level C — Container / orchestration security (behavioral proxies — no real exploit)

ID	Fault	Target	How to simulate	Source / reference
C1	**Privileged / misconfigured container behavior**	any NF pod	Deploy NF variant with excess privileges doing anomalous host-ish syscalls; capture via eBPF (Beyla/Falco).	**Aiello**: container runtime escape 0.85 (Critical, #1 risk). Proxy only — no real escape CVE executed.
C2	**Malicious sidecar / rogue in-pod process**	NF pod	Inject a benign extra process (CPU/net/file activity) into an NF container mid-run.	**Aiello**: container security (Sec. 4.1). Proxy for post-compromise behavior; visible in cAdvisor + eBPF.
C3	**K8s API / RBAC abuse signal**	control plane	Trigger unauthorized-ish K8s API / ServiceAccount-token use in a test namespace.	**Aiello**: cites CVE-2023-2727 (K8s SA tokens). Maps to K8s audit/events layer.
C4	**Config / secret tampering**	NF configmap/secret	Mutate NF config (bad NRF URL, wrong cert) and restart.	**Thesis**: overlaps environment/config faults (Zhou taxonomy). **Aiello**: misconfiguration theme.

Level D — Dependency & data-layer attacks

ID	Fault	Target	How to simulate	Source / reference
D1	**NRF poisoning / rogue registration**	NRF	Register a fake/malformed NF profile via the NRF SBI API.	**Aiello**: SBA API authorization under-deployed. Extends **Thesis** NRF dependency class (interaction fault).
D2	**MongoDB auth failure / tampering**	MongoDB / UDR	Revoke DB creds or corrupt a subscriber record mid-run.	**Thesis**: extends the MongoDB pod-kill dependency fault into a security variant.
D3	**SBI / SCP unauthorized access or TLS failure**	SCP, any NF	Disable/expire TLS on an SBI link, or inject unauthorized SBI calls.	**Aiello**: SBA API authorization strengthening (Sec. 5). **Thesis**: SCP is the model-D hub.

Level E — Baseline operational faults (kept from thesis, trimmed to ~6–8 anchors)

ID	Fault	Target	How to simulate	Source / reference
E1	**CPU stress**	AMF, SCP	Chaos Mesh StressChaos/cpu (2 workers @ 80%).	**Thesis**: CPU-stress class (faults 01, 08).
E2	**Memory pressure / OOM**	UPF, AMF	Chaos Mesh StressChaos/mem.	**Thesis**: memory-pressure class (faults 02, 22).
E3	**Pod crash**	AMF, SMF	Chaos Mesh PodChaos/pod-kill.	**Thesis**: pod-crash class (faults 03, 07).
E4	**Network delay**	SCP	Chaos Mesh NetworkChaos/delay (500 ms).	**Thesis**: network-delay class (fault 17).
E5	**Packet loss**	UPF	Chaos Mesh NetworkChaos/loss (30%).	**Thesis**: packet-loss class (fault 06).
E6	**NRF / MongoDB kill**	NRF, MongoDB	Chaos Mesh PodChaos/pod-kill.	**Thesis**: dependency-failure class (faults 15, 20).

Tooling summary: Chaos Mesh (E, C1–C2, D2) • UERANSIM at scale (A1, A3) • 5GReplay / Montimage (A2, A3, B4) • pfcpsim / ONF (B1, B2) • public PoC scripts + CVEs (B2, B3) • scapy/custom gRPC (B4) • Falco/eBPF-Beyla to observe Level C.
Two open decisions: (1) keep Level C as safe behavioral proxies or drop it; (2) consider adding a security/audit modality (K8s audit events + eBPF/Falco alerts) — the natural 'new modality' that strengthens the extension claim, mirroring AnoMod adding CCR/API-Res.